

Low-Power Retentive True Single-Phase-Clocked Flip-Flop With Redundant-Recharge-Free Operation

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Objectives:

- To implement a flip-flop that eliminates "redundant pre-charge" operations to minimize power consumption in low-data-activity digital systems.

Status of the work done:

- Schematic design and transistor sizing completed based on the proposed architecture, and waveforms generated.

Tools Used:

- Cadence Virtuoso

Problem Statement:

- Goal: To create a circuit that only pre-charges when a state change is necessary, significantly boosting efficiency at low data activity (typically for IoT devices).

Architecture	Key Drawbacks & Power Inefficiency
XCFF / ACFF	Current Contention: Strong contention at output nodes; wasted power during precharge/discharge.
TCFF	Voltage Drops: High transistor sharing causes internal voltage drops; unsuitable for low-voltage operation.
SPC-18T	Robustness: Redundant internal node switching; high hold-time requirements (requires extra buffers).
S2CFF	Redundancy: Successfully operates at low voltage, but still suffers from redundant precharge/discharge power waste.
RTFF	Circuit Flaws: Short-circuit paths and NMOS pull-up usage lead to internal voltage drops and poor robustness.

Proposed Circuit and Sizings

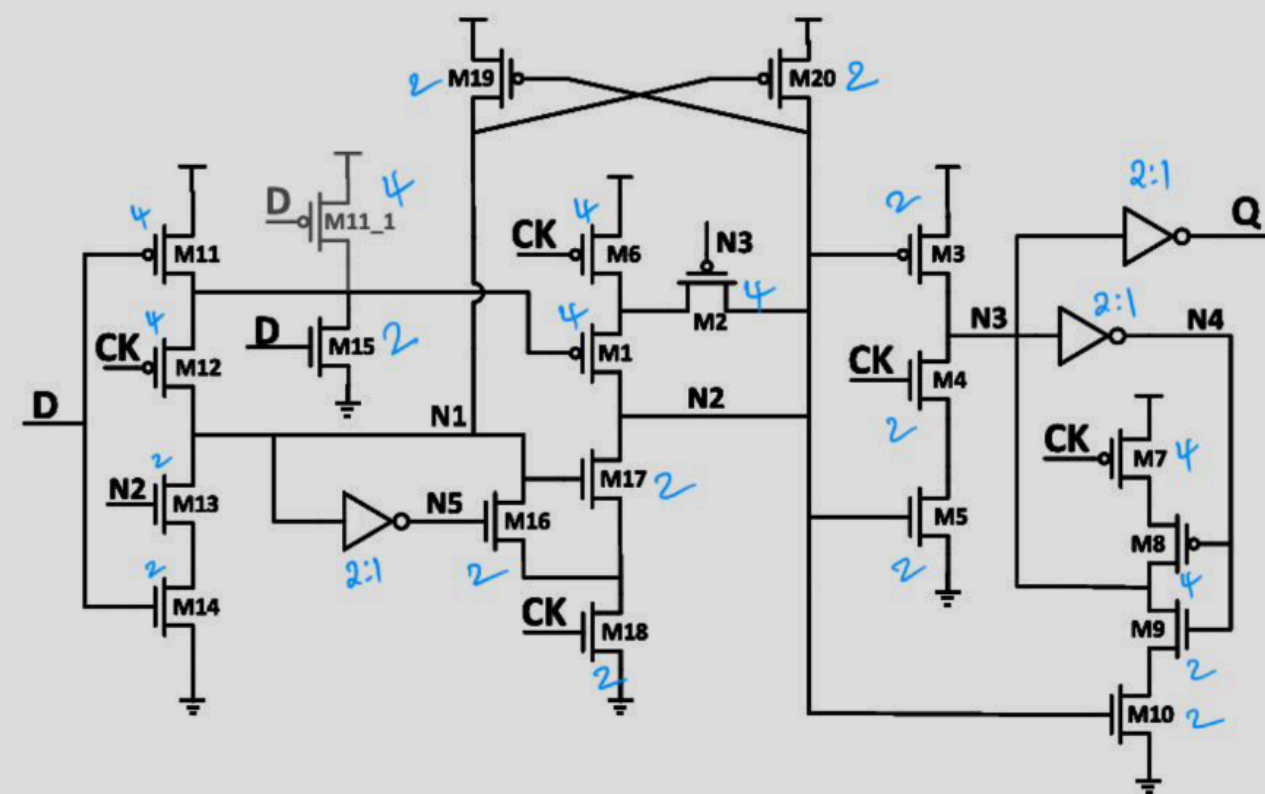


Fig. 4. Schematic of the proposed FF.

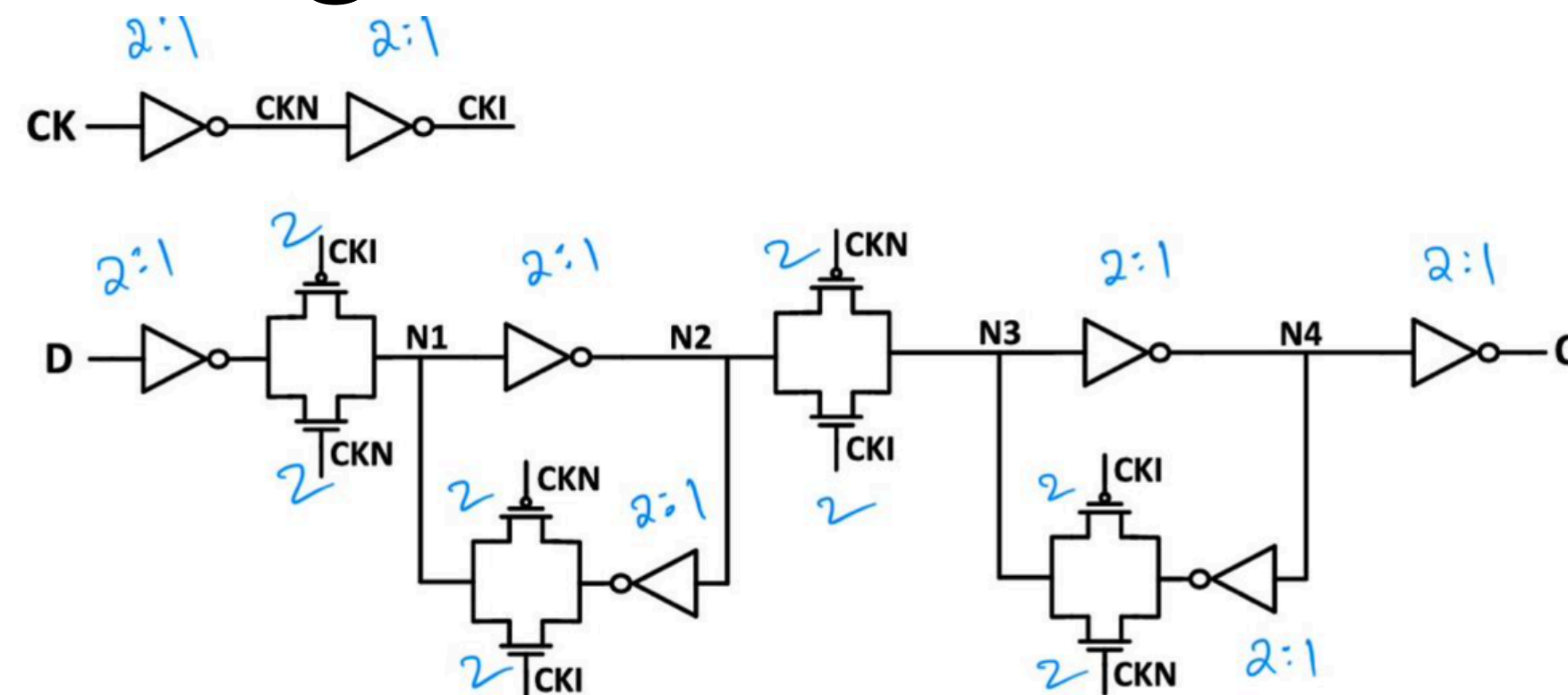


Fig. 1. Schematic of TGFF.

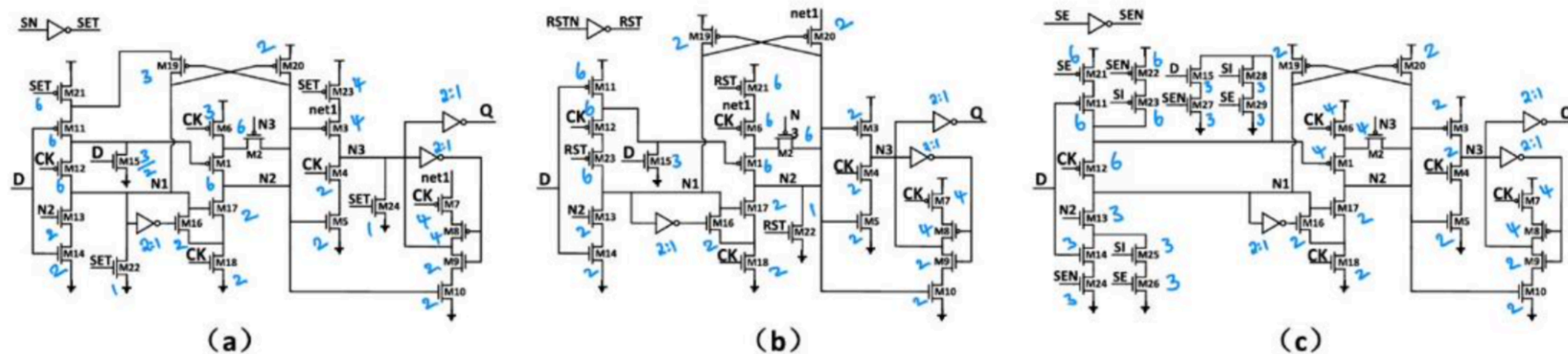


Fig. 7. Schematic of the proposed FF with (a) set, (b) reset, and (c) scan.

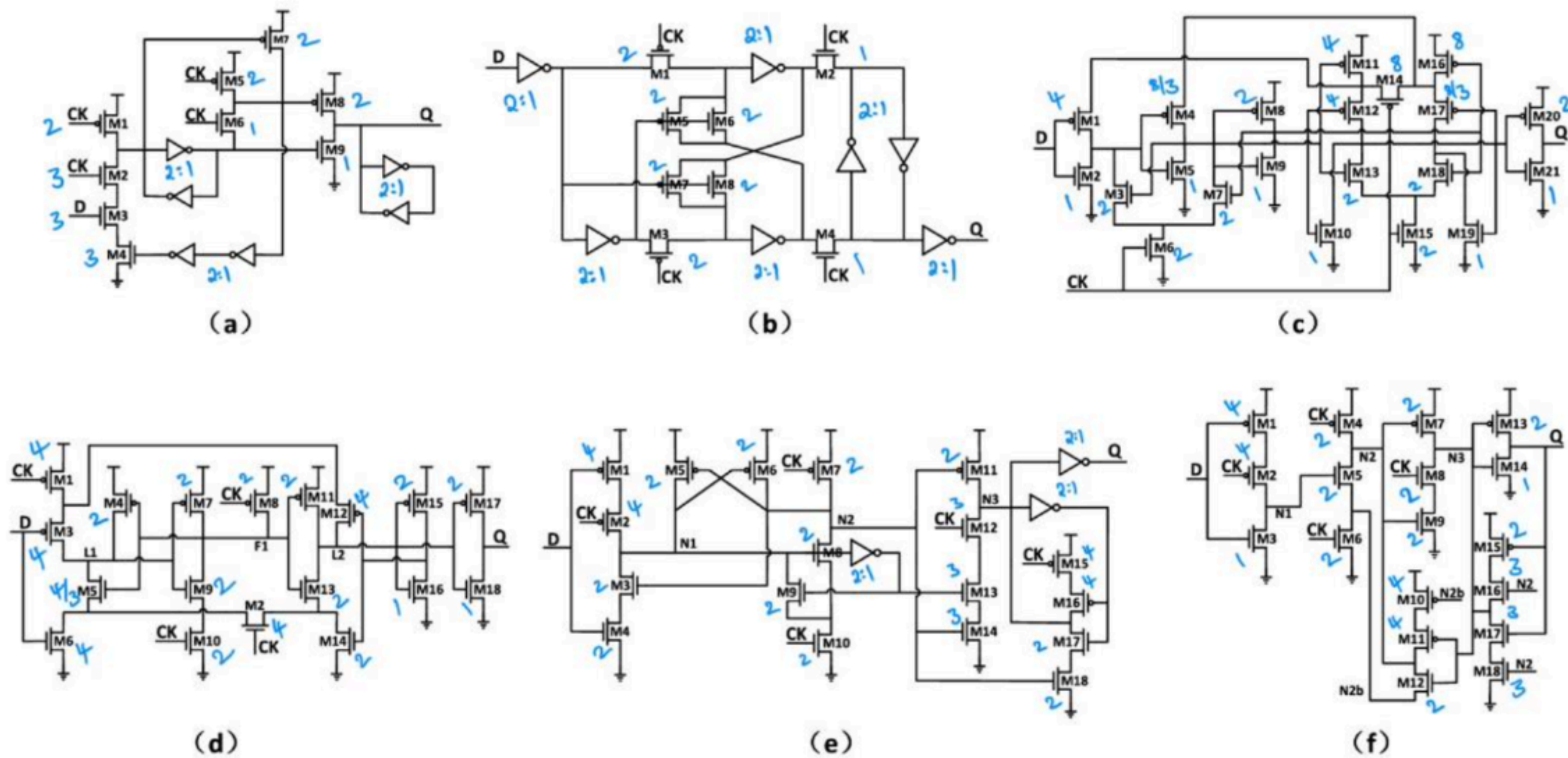
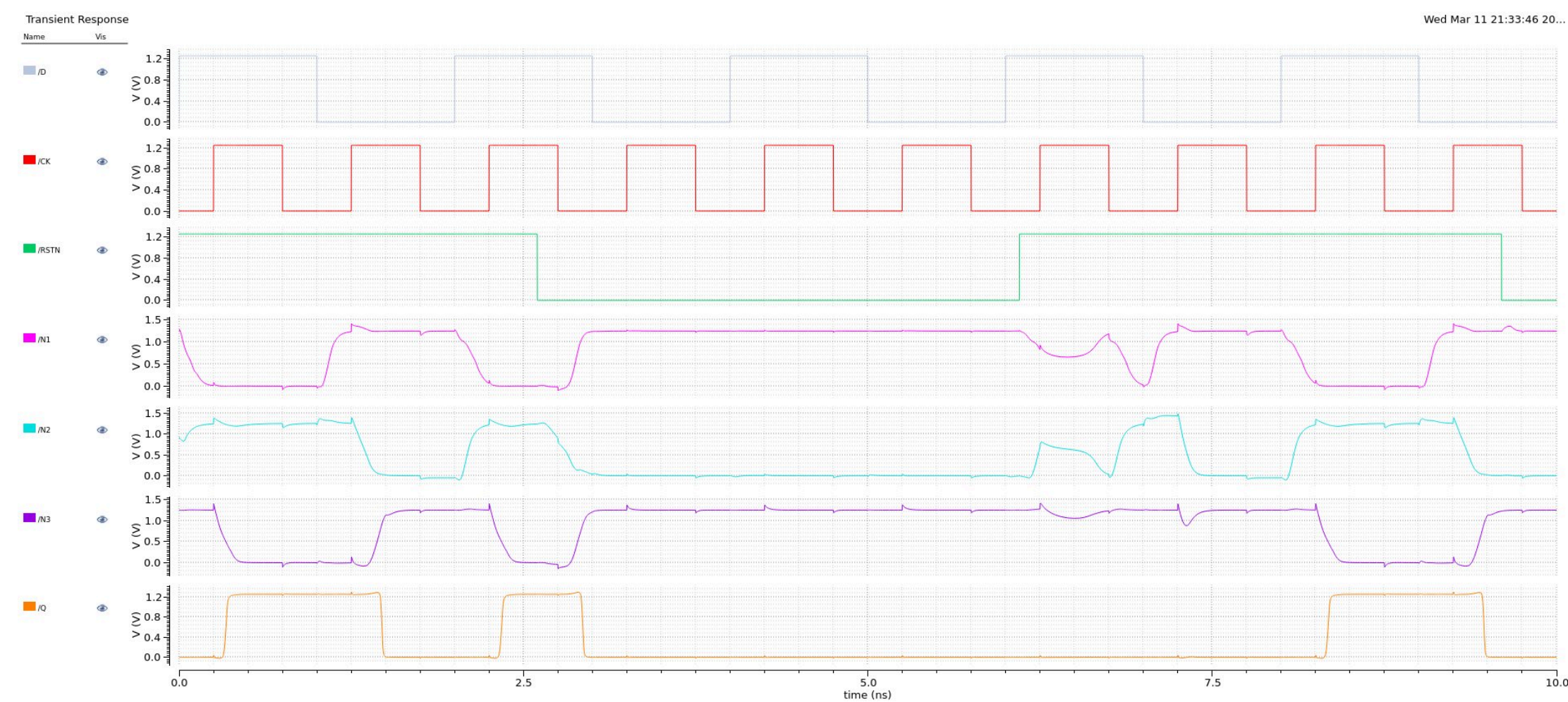
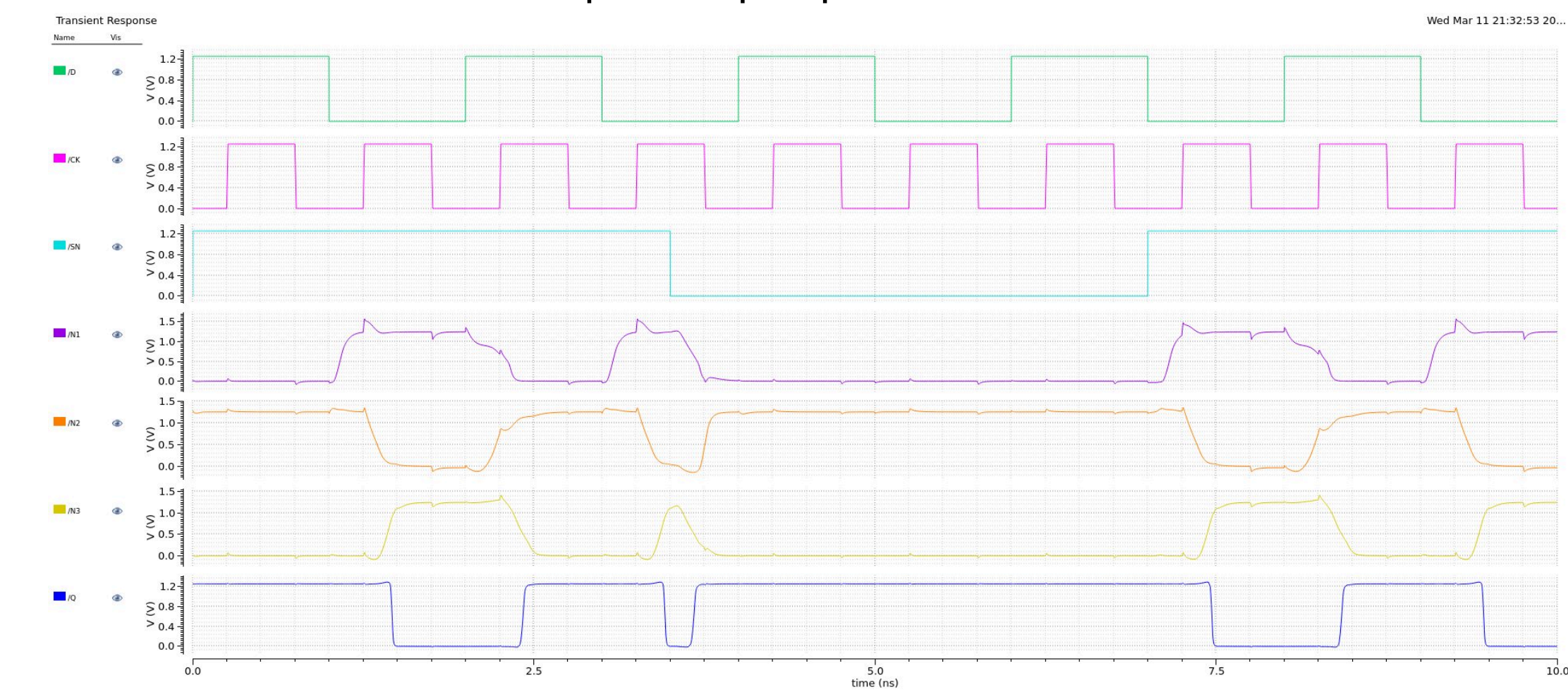


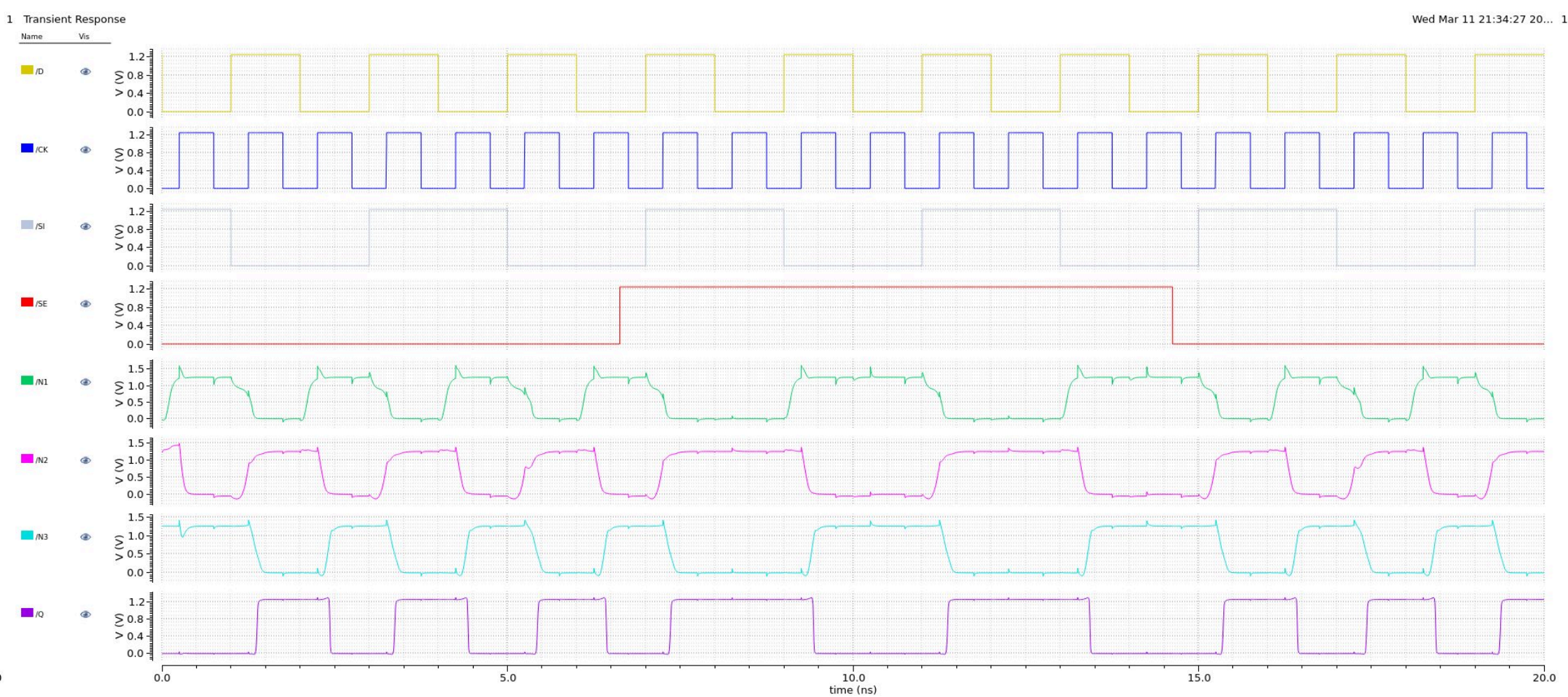
Fig. 2. Schematic of various FFs. (a) XCFF [13]. (b) ACFF [14]. (c) TCFF [15]. (d) SPC-18T FF [16]. (e) S^2 CFF [17]. (f) RTFF [18].



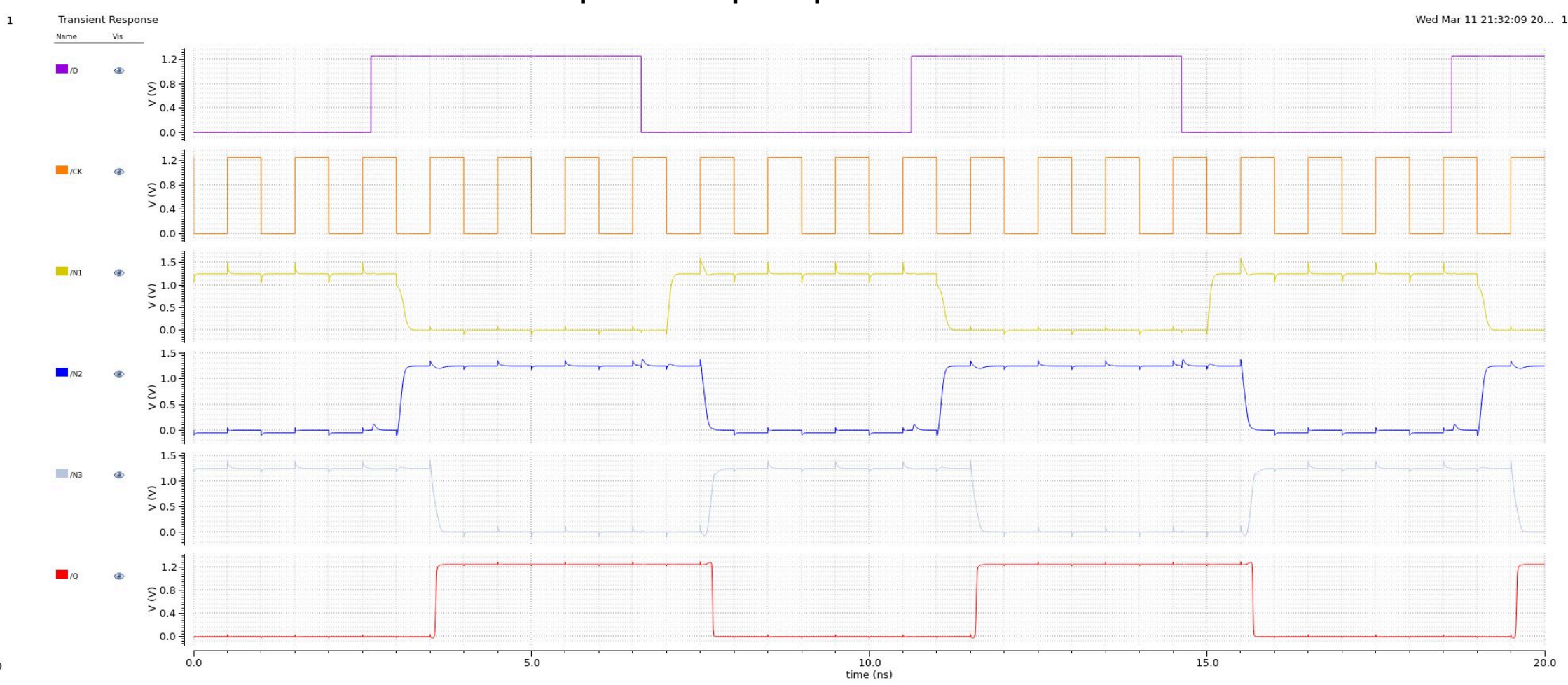
Proposed Flip Flop with Reset



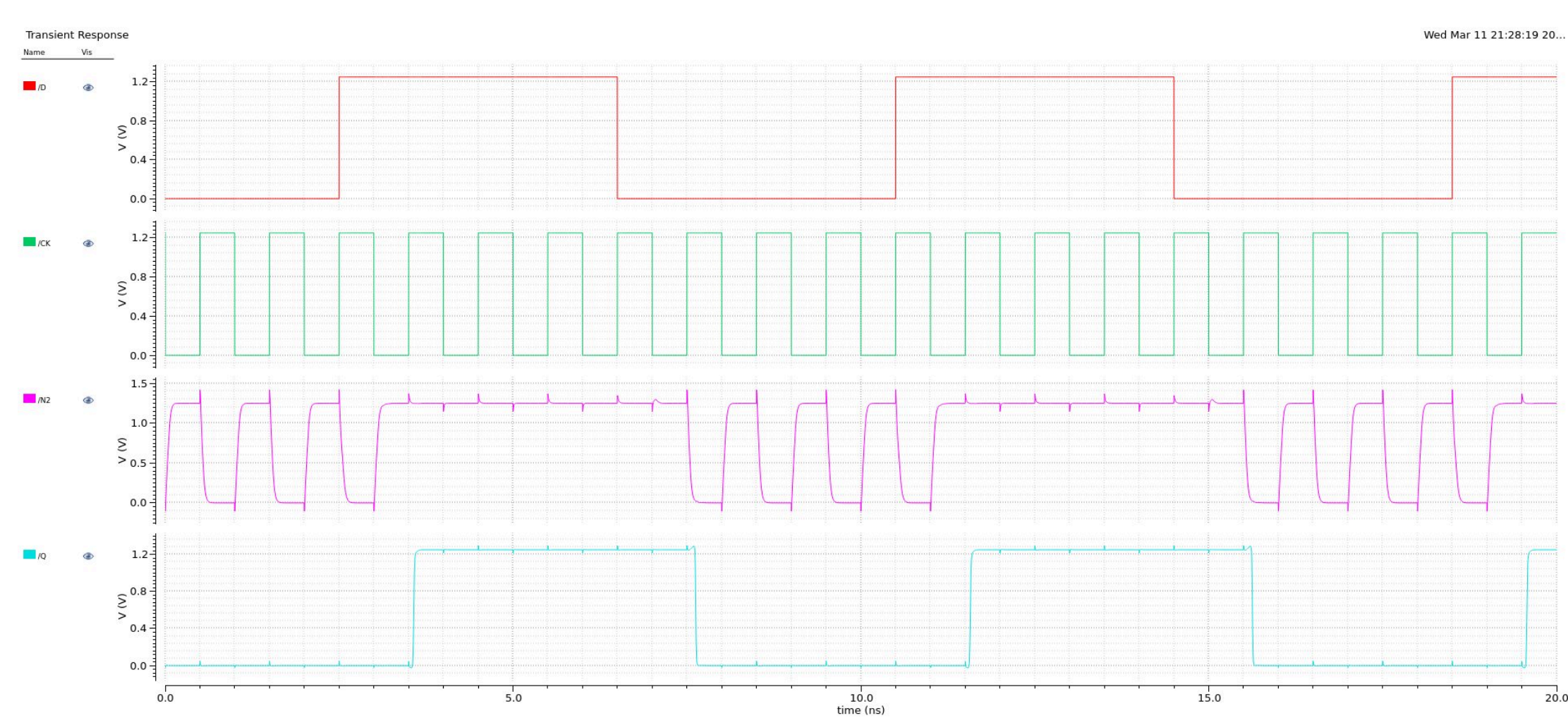
Proposed Flip Flop with Set



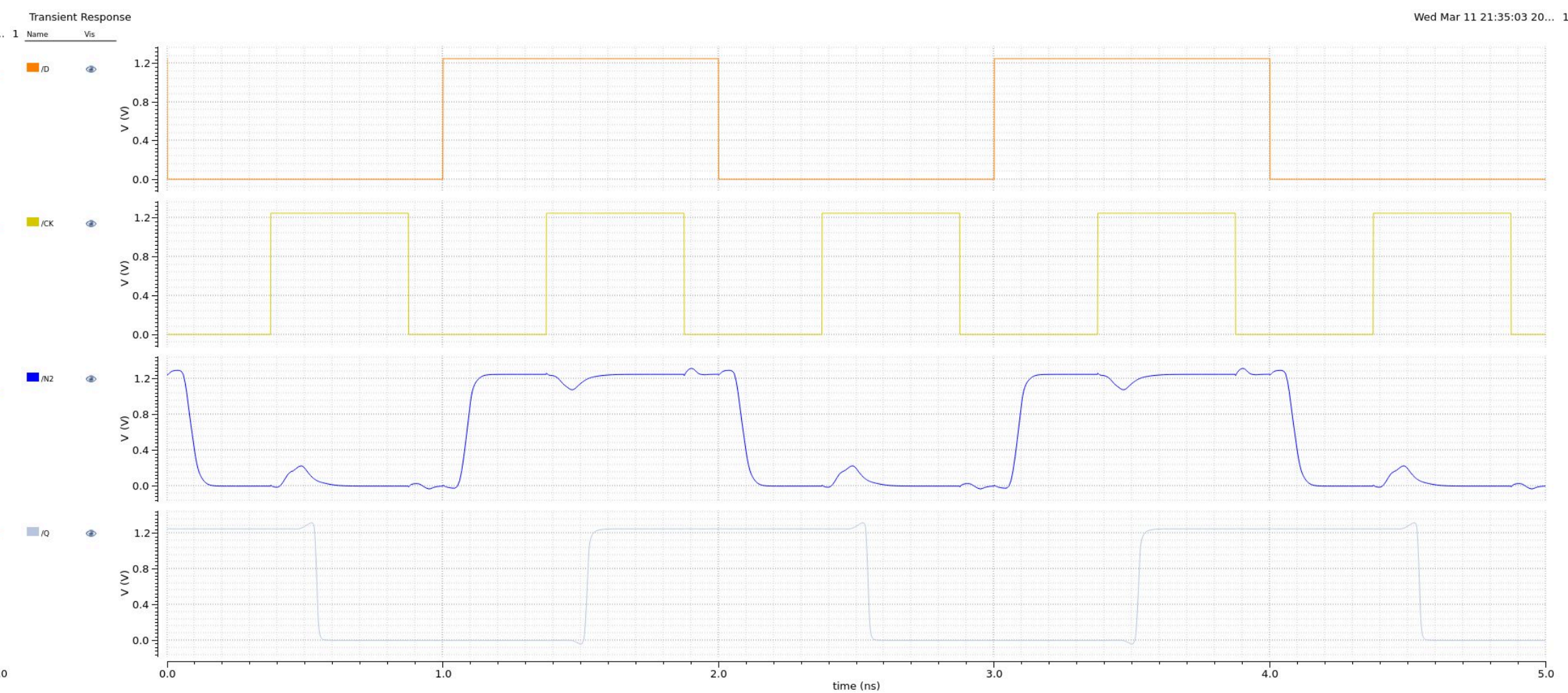
Proposed Flip Flop with Scan



Proposed Flip Flop



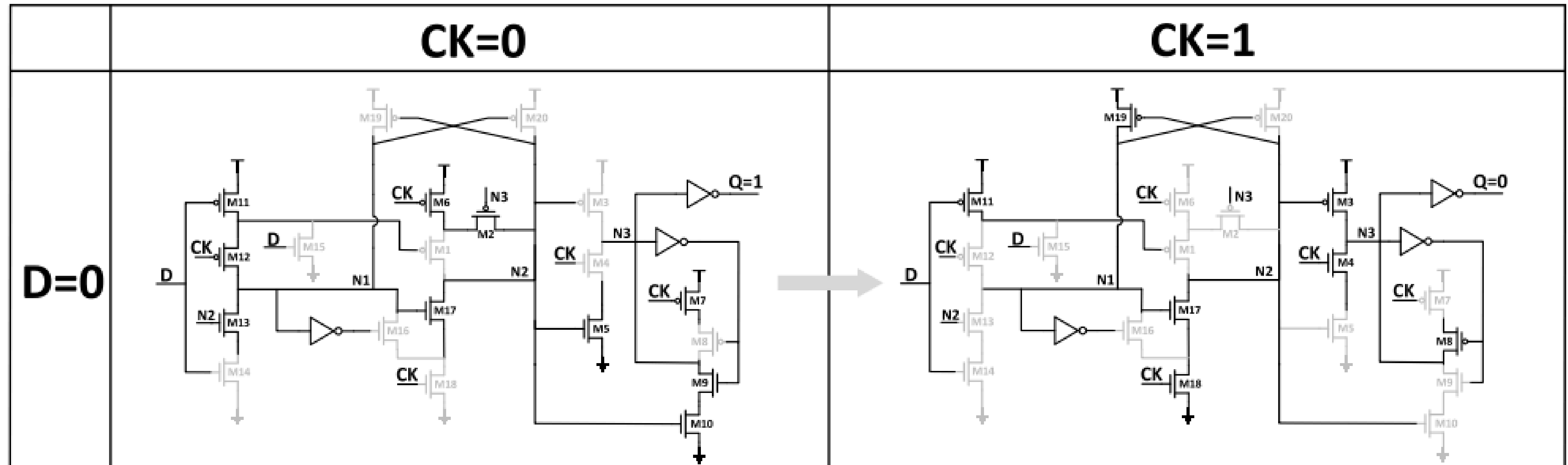
S²-CFF



TG-FF

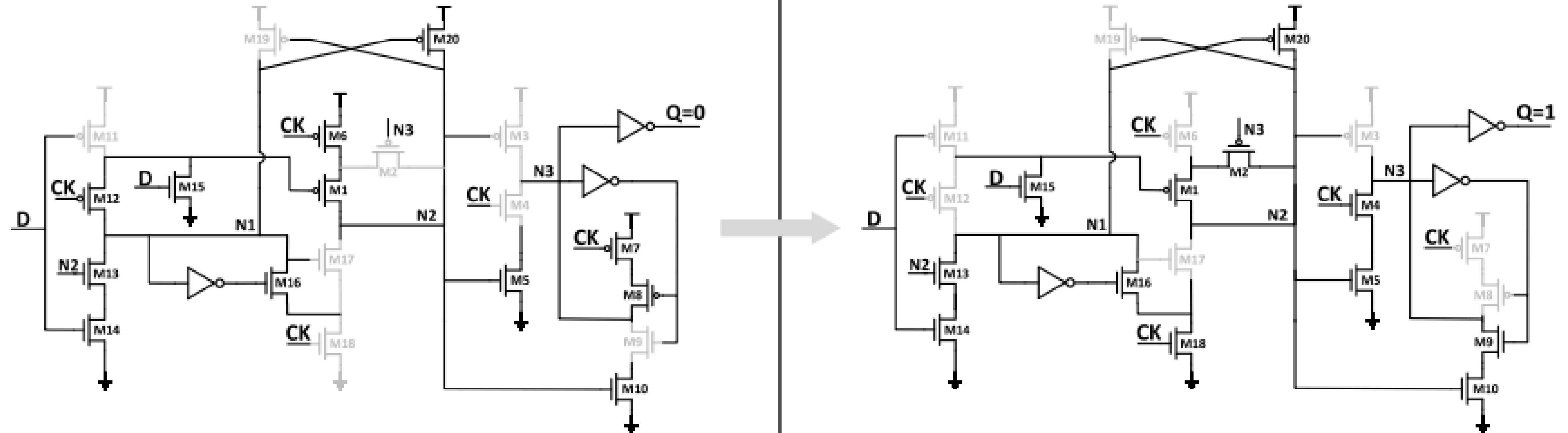
After Mid Progress

Proposed FF working for D = 0



Proposed FF working for D = 1

D=1



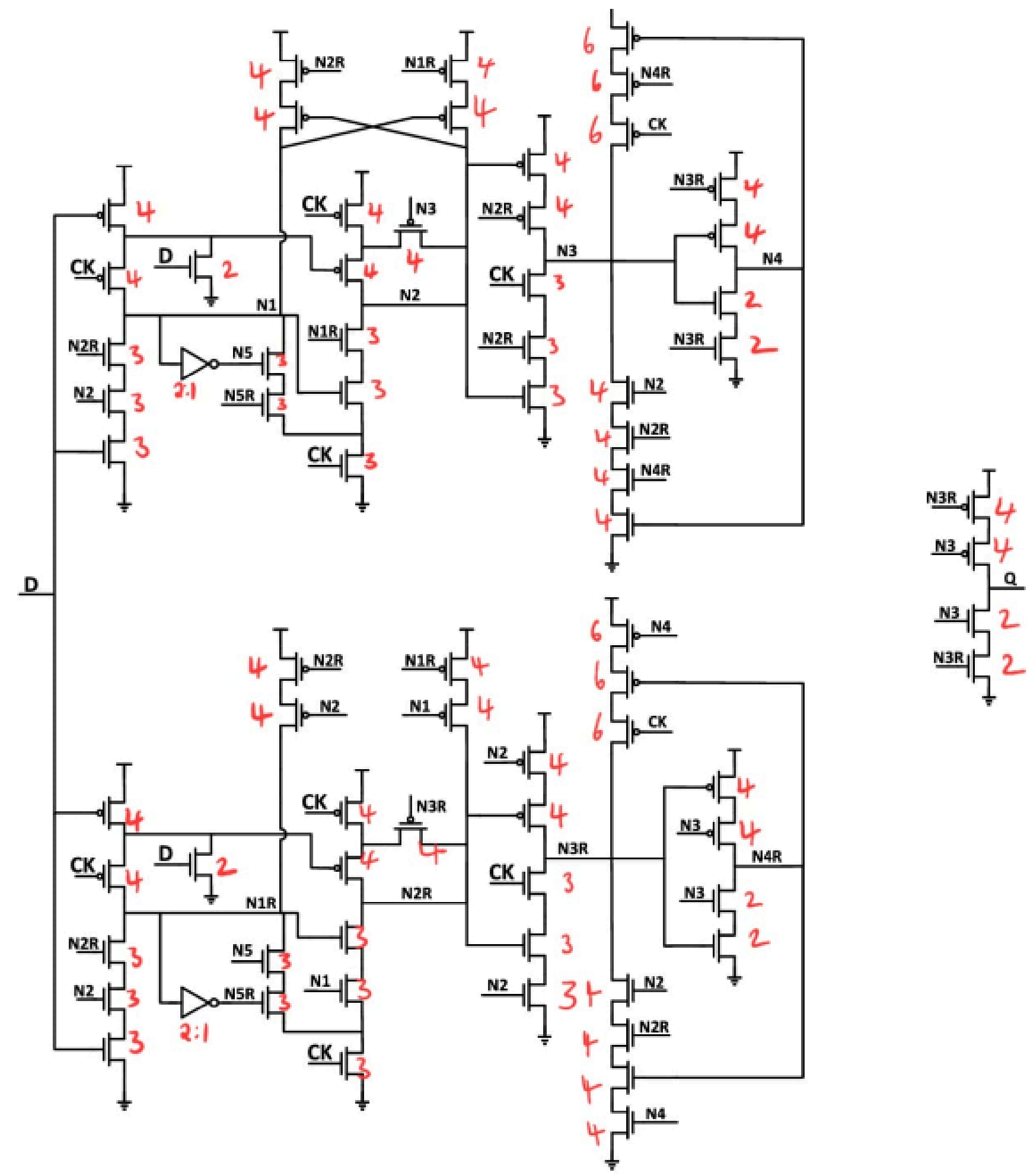
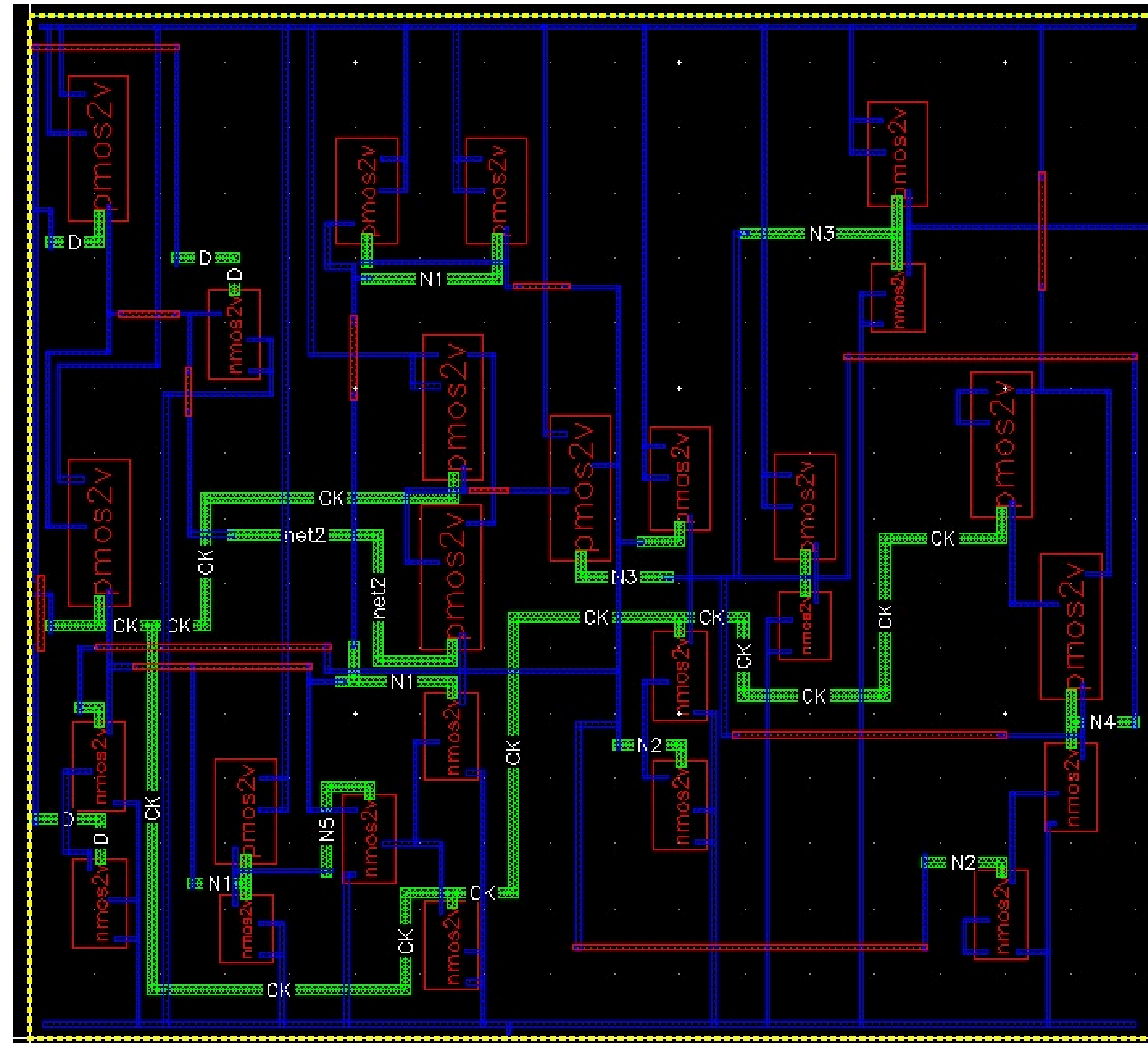
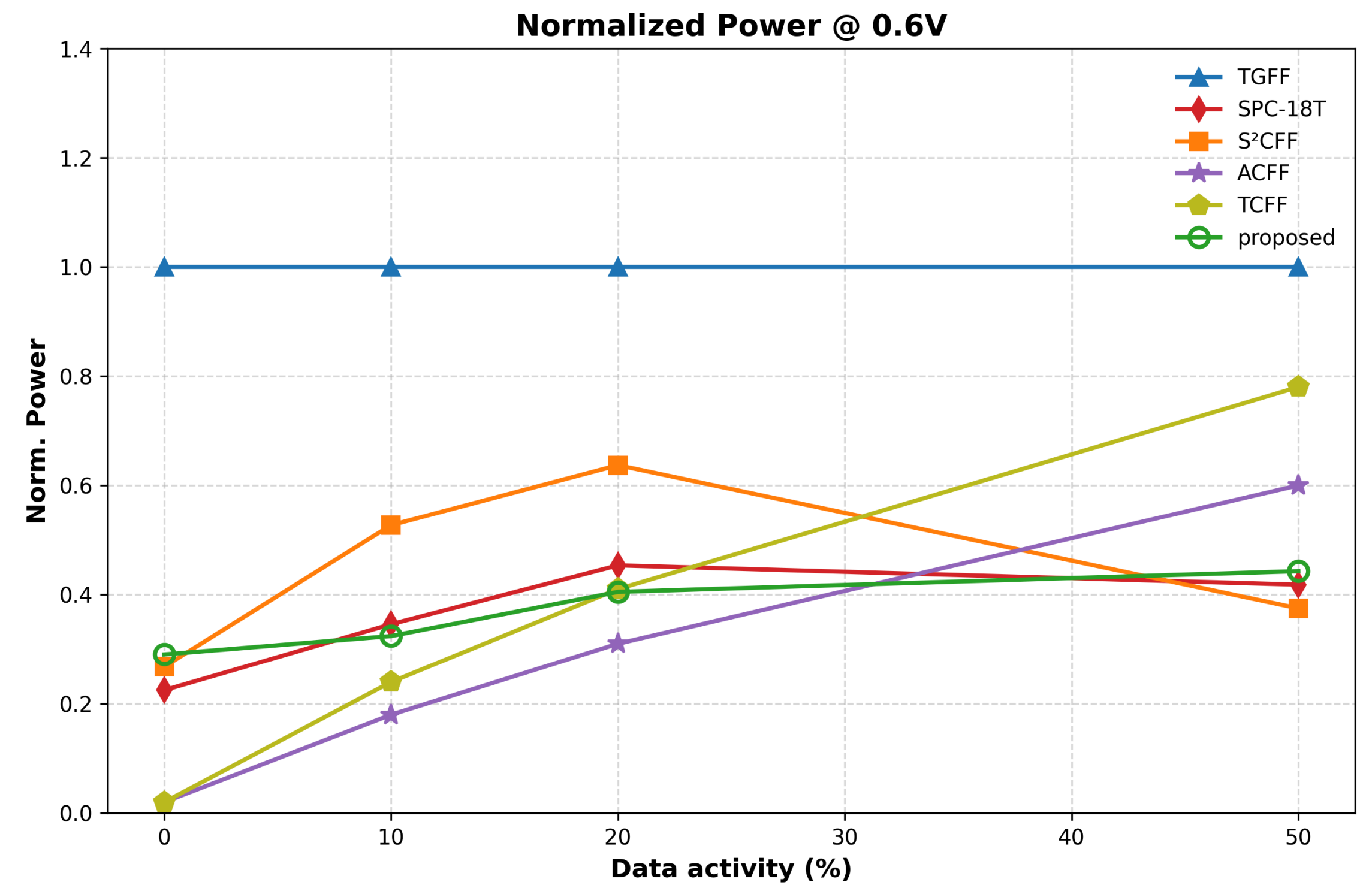
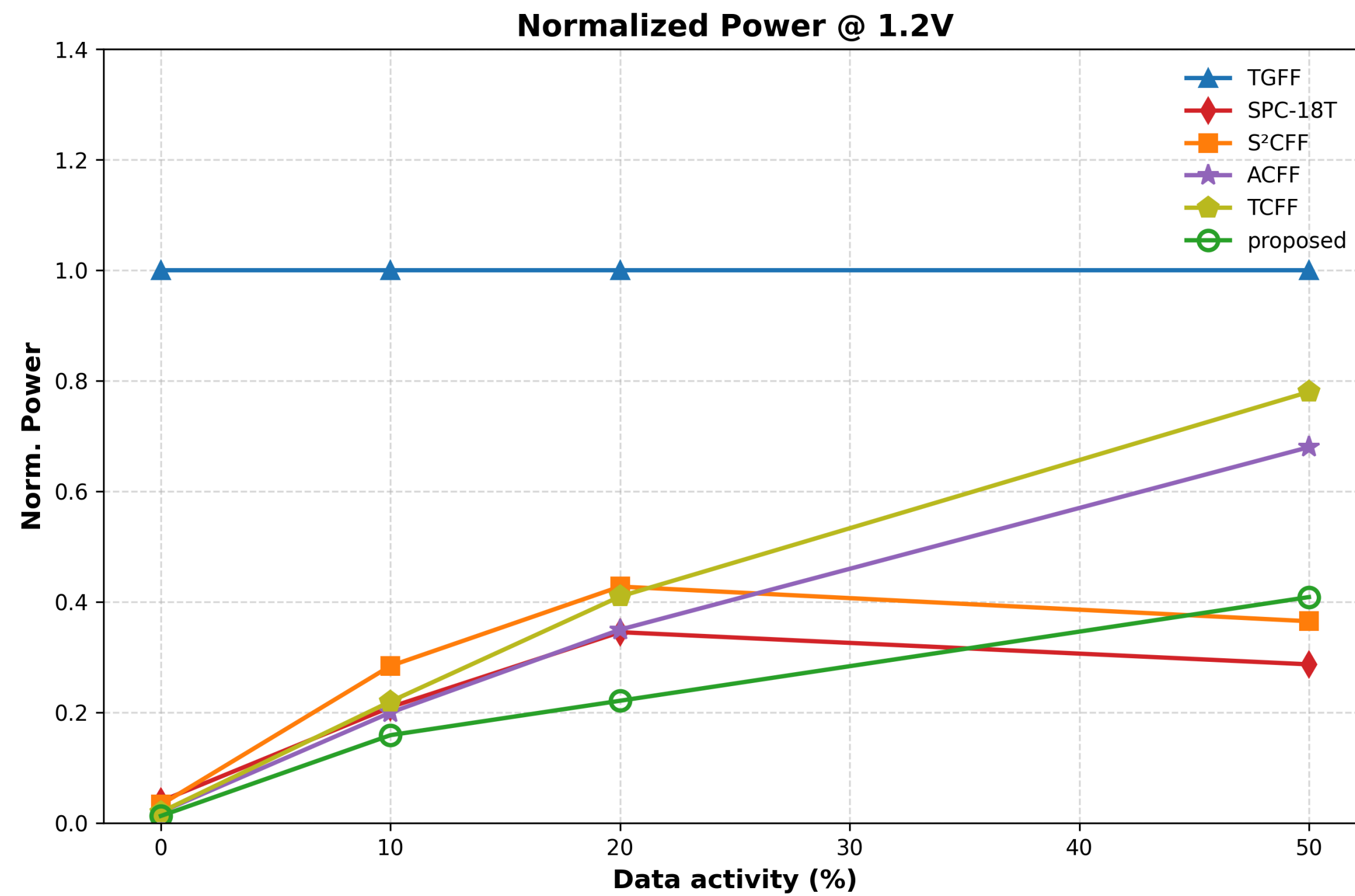


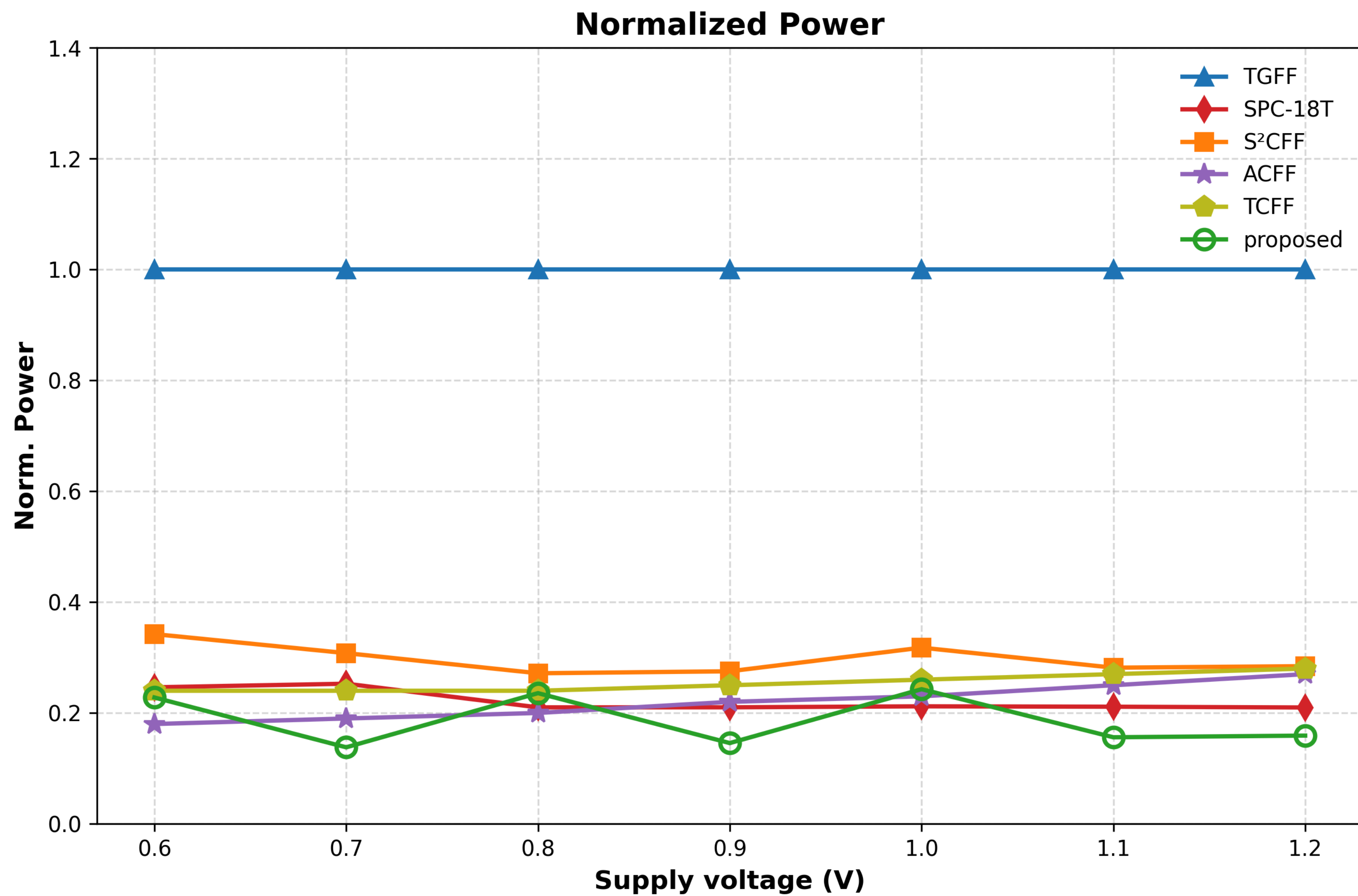
Fig. 10. Schematic of the proposed SEU-tolerant FF.

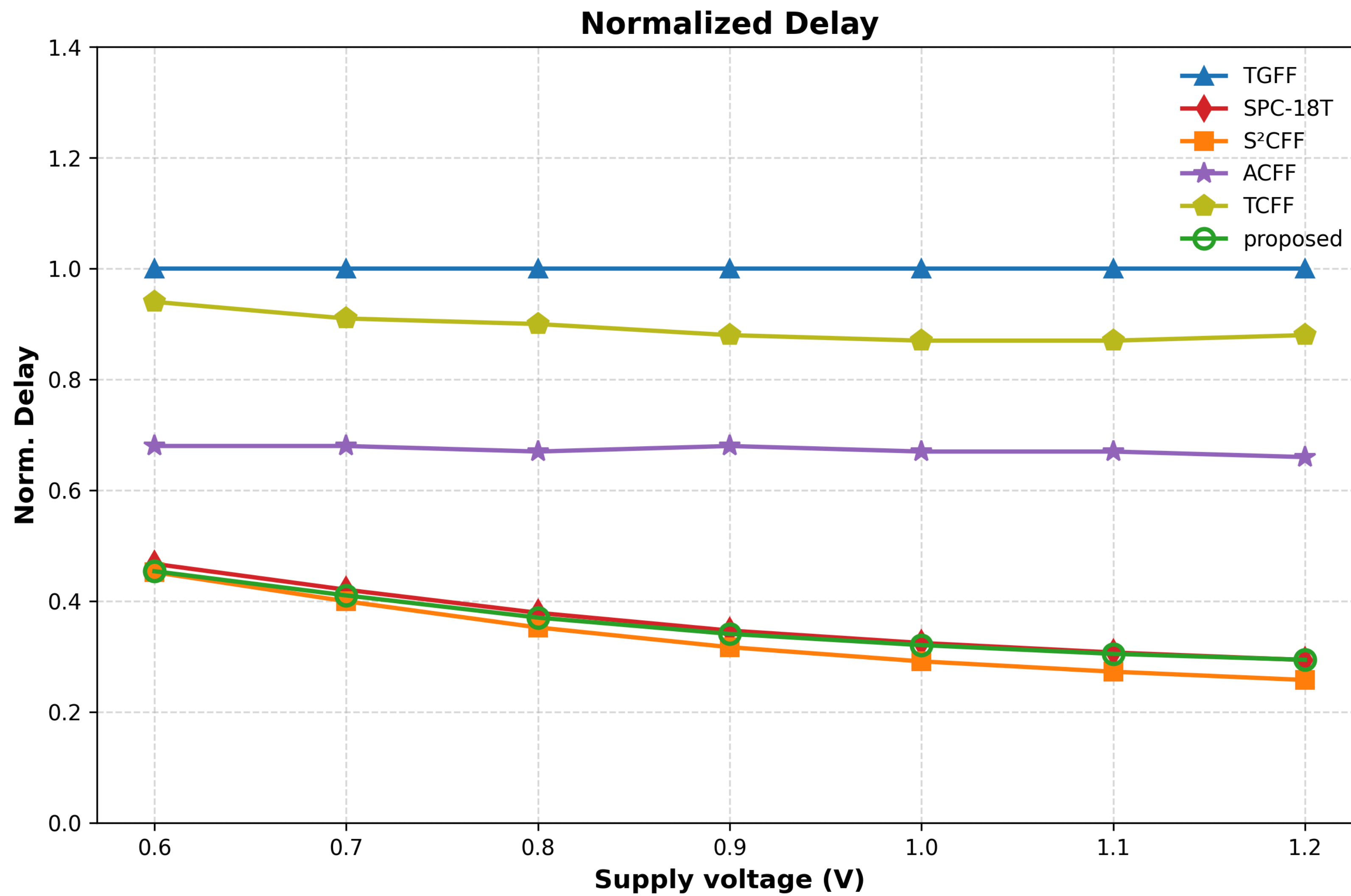
Layout of the Proposed FF

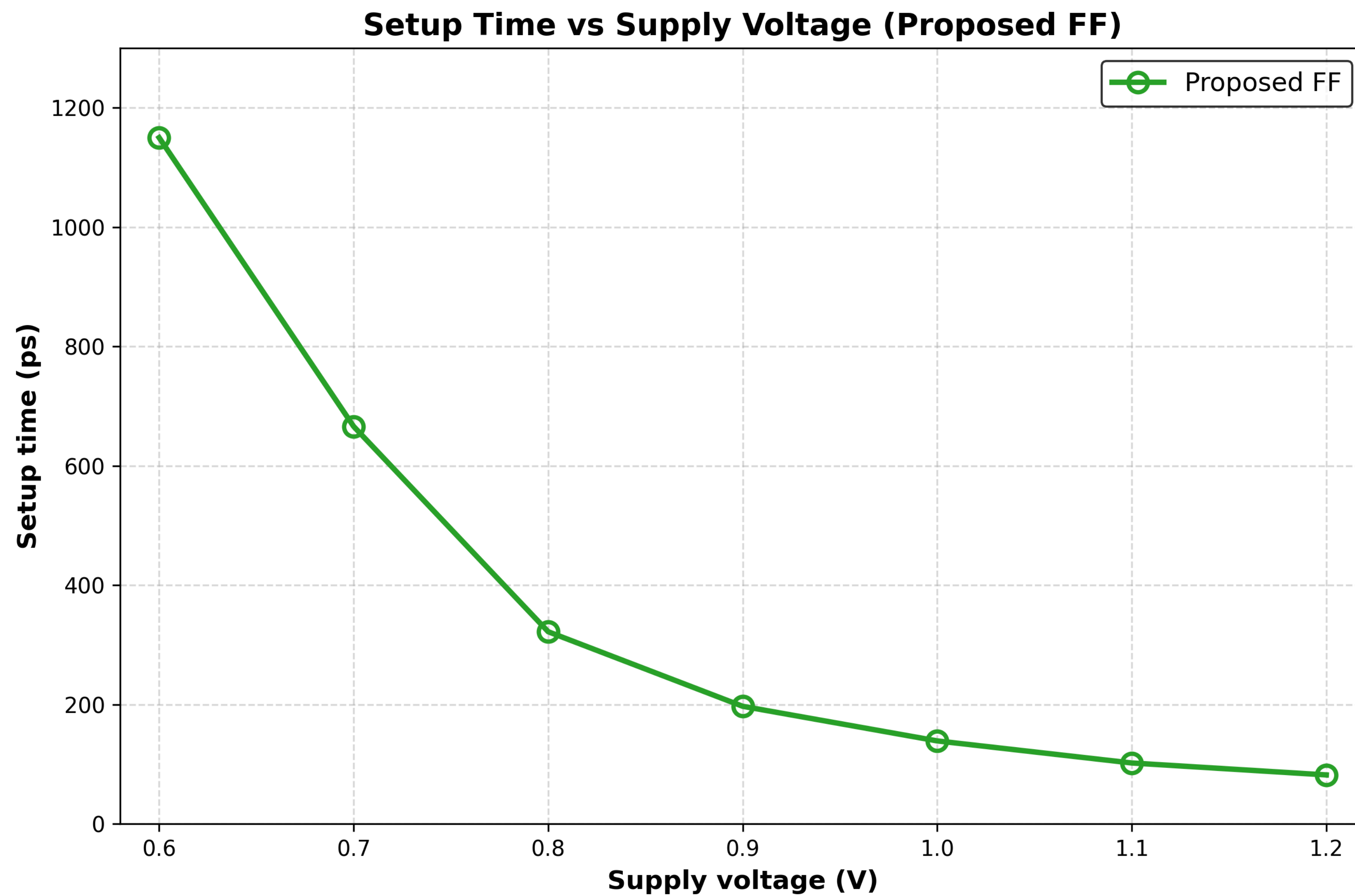


Virtuoso Simulation Results (45nm Technology Node)









Setup Time(in picoseconds)

TGFF	SPC-18T	S2-CFF	ACFF	Proposed	P-SET	P-RST	P-SCAN
10	10	10	240	82	112	196	224

Thank-You!